

DNA Isolation and DNA Spectrophotometry

Laboratory Report — Concentration and Purity Assessment

Analyst: Gontse Bridgett Baloyi
Technique category: Molecular Biology

1. Objective

To calculate DNA concentration and assess purity (A260/A280 and A260/A230 ratios) for three isolated DNA samples using UV-Vis spectrophotometry.

2. Introduction

Once DNA has been isolated and purified, spectrophotometry provides a fast, non-destructive way to determine both yield and purity. Nucleic acids absorb UV light maximally at 260 nm, allowing concentration to be calculated directly from absorbance. Two ratios are used to assess purity: A260/A280, which flags protein contamination (proteins absorb near 280 nm), and A260/A230, which flags contamination from residual solvents, guanidine salts, or carbohydrates carried over from extraction (these absorb near 230 nm). Both ratios should fall within expected ranges for DNA to be considered suitable for sensitive downstream applications such as PCR or sequencing.

3. Materials

- Isolated DNA samples
- TE buffer or nuclease-free water (dilution/blank)
- UV-Vis spectrophotometer capable of A230, A260, A280 readings

4. Method

- Blanked the spectrophotometer using elution buffer/water.
- Loaded 1–2 uL of each DNA sample onto the spectrophotometer.
- Recorded absorbance at 230 nm, 260 nm, and 280 nm for each sample.
- Calculated DNA concentration from A260 using the standard dsDNA conversion factor (50 ug/mL per A260 unit).
- Calculated A260/A280 and A260/A230 ratios and assessed against expected purity ranges.

5. Results

Concentration and purity varied across the three samples. Sample_1 met both acceptable purity ranges; Sample_2 fell outside both ranges; Sample_3 showed an elevated A260/A280 ratio despite an acceptable A260/A230 ratio.

Sample	Conc. (ug/mL)	A260/A280	A260/A230	Status
Sample_1	726.0	1.83	2.15	Pure

Sample_2	443.0	1.52	1.40	Protein & solvent contamination
Sample_3	1052.0	2.15	2.05	Possible RNA contamination

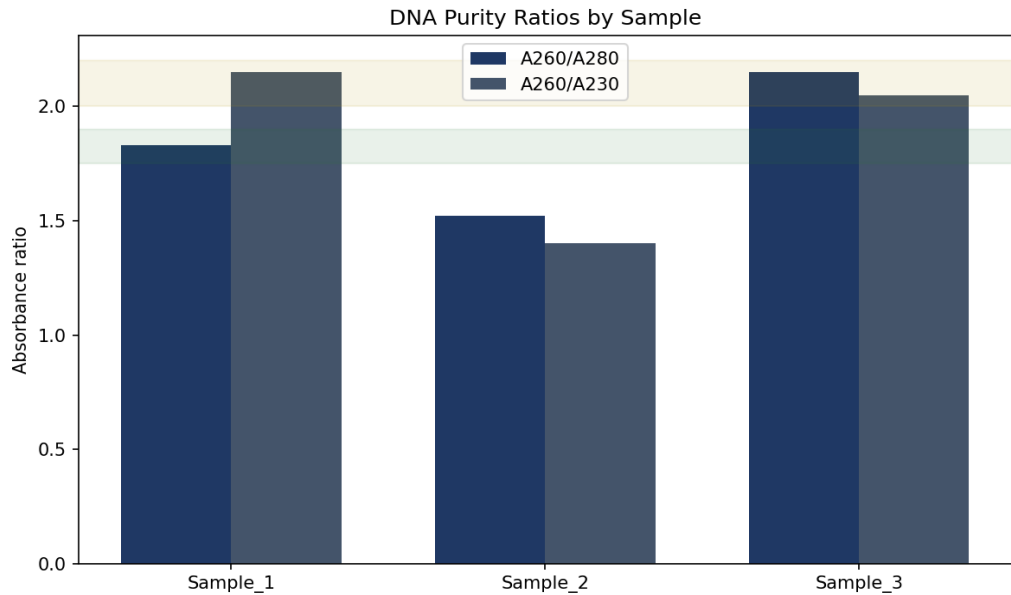


Figure 1. A260/A280 and A260/A230 ratios by sample, with acceptable ranges shaded.

6. Discussion

Sample_2's low values on both ratios point to two distinct contamination sources acting together: a low A260/A280 (below the 1.75 threshold) indicates residual protein was not fully removed during purification, while a low A260/A230 indicates residual extraction solvent, salts, or carbohydrates were also retained — suggesting the purification step for this sample was generally incomplete rather than a single isolated failure. Sample_3's elevated A260/A280 ratio (above 1.90), combined with an acceptable A260/A230, is a more specific signature: RNA co-purified alongside the DNA, since RNA also absorbs near 260 nm and elevates that ratio without affecting the 230 nm-associated contaminants. In practice, Sample_2 would require re-purification (repeat protein removal and solvent/salt washing steps), while Sample_3 would benefit from RNase treatment before use in RNA-sensitive downstream applications.

7. Conclusion

This exercise demonstrated how the two standard DNA purity ratios provide complementary, not redundant, information — together they can distinguish between protein contamination, solvent/salt contamination, and RNA contamination, each of which would require a different corrective step before the DNA is suitable for sensitive downstream applications.